BRAC University **Department of Computer Science and Engineering** Semester: **Fall 2024** Course Code: **CSE460** Course Title: **VLSI Design**	**Midterm Examination** Full Marks: **10 x 2 + 15 = 35** Time: **1 hour 15 minutes** Date: **22 November 2024**

Set A

Student ID:	**Name:**	**Section:**

[Answer all **THREE** questions. Return the question paper with the answer script.]

1. (CO2) Consider a sequential circuit that detects if two consecutive bits are the same or not. The circuit generates an output z=1 if two consecutive bits are opposite and z=0 if the bits are the same. The output changes in the same clock cycle where the input is received.

w	0	**0**	**1**	**0**	**0**	**1**	**1**	**0**
z	0	0	**1**	**1**	0	**1**	0	**1**

Consider overlapping bit sequence

(a) Determine the type of FSM required for solving this problem. **1**

(b) Draw the state diagram of the circuit. **3**

(c) Derive a state-assigned table from (b) using the gray-encoding technique. **3**

(d) Find the next-state and output logic expressions using k-map. **3**

2. (CO2) Consider a sequential circuit with two inputs (p1 & p2) and a single output, z. The decisions regarding the status of various product orders are handled by two departments: the Sales Team and the Warehouse. Both departments independently approve (denoted by 1) or reject (denoted by 0) an order. The decisions of the Sales Team are given by a bit stream, p1, while the decisions of the Warehouse are represented by p2. If, for two consecutive orders, both departments approve (**i.e., p1=p2=1 for two consecutive clock cycles**), the sequential circuit provides **z=1 at the next order**. The decisions of the Sales Team & the Warehouse for several orders and their corresponding synchronization instances are given below:

p1	0	**1**	**1**	**1**	**1**	1	1	0	0	0	1
p2	1	**1**	**1**	**1**	**1**	0	1	0	0	1	1
z	0	0	0	**1**	0	**1**	0	0	0	0	0

Note:

- **You must use binary encoding to assign states.**
- **The overlapping of the input bits (p1 & p2) is not allowed.**

(a) Determine the type of FSM that would be needed for the given problem? **1**

(b) Design the state diagram for the corresponding FSM **2**

(c) Derive the state table and state-assigned table from your state diagram in (b). **3**

(d) Evaluate the next state & output logic expressions to implement the FSM & calculate the number of flip-flops required for the circuit implementation. (You don't need to draw the circuit implementation). **4**

3. (CO1) We have to implement an exam cheat detector. We have 4 sensors (A, B, C). Their functions are as follows:

A: High when a person has AirPods hidden in his/her ears.
B: High when a person has something written on his/her arms.
C: High when a person has a cheatsheet in his/her pocket

The output of the system will be high if the student should be expelled. The system is lenient enough to not expel if a student commits only offense **A** or only offense **B**. In any other case, the student will be expelled.

(a) **Write** down the truth table for the system, describing all possible states of the sensors (A, B, C) and their corresponding output states. **2**

(b) **Draw** the Karnaugh map (K-map) for the system based on the truth table in part (a). Derive the logical expression that governs the system from the K-map. **4**

(c) **Draw** the CMOS logic circuit for the logical expression derived in part (b). Specify the number of MOSFETs required to implement your design. **3**

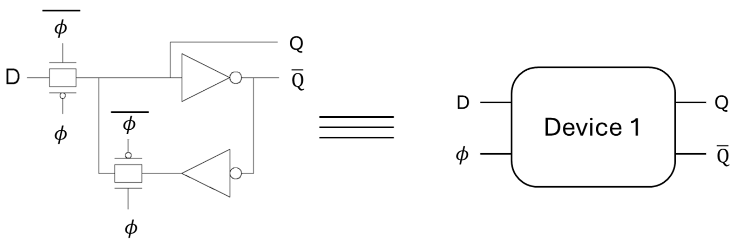

Figure 1: Circuit on the left is represented by the block - **Device 1**

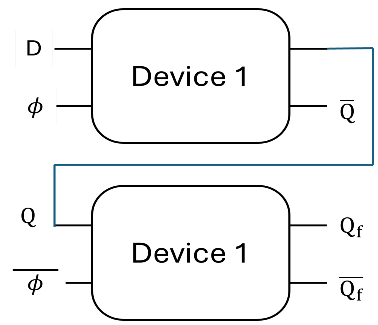

Figure 2: A circuit made up of two **Device 1** blocks **from Fig. 1**

(d) **Analyze** the circuit in **Fig. 1** and write the truth table of the block **Device 1**. **Represent Q as a logic function of D when $\phi = 0$.** **3**

(e) The table represents the time-dependent logical values (0 or 1) for each node in the circuit depicted in **Fig. 2**, which is constructed using two Device 1 blocks**.** The Φ and D values are provided in the table. **3**

Node ↓ / Time →	5 ns	10 ns	15 ns	20 ns	25 ns
ϕ	0	1	0	1	0
D	0	1	1	1	0
Q					
Q_f	1				

Determine the values of **Q** and Q_f for the specified time intervals (5 ns, 10 ns, etc.). **Determine** if the circuit in **Fig. 2** functions as a **negative**-edge/level-triggered or **positive**-edge/level-triggered flip-flop/latch?